Namala Krishna Chaitanya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.17
FIITJEE JUNIOR COLLEGE	Class XII (TELANGANA BOARD OF INTERMEDIATE EDUCATION)	2024	98.60%
PANCHAVATI VIDYALAYA	Class X (TELANGANA BOARD OF SECONDARY EDUCATION)	2022	10.00%

SKILLS

Programming Languages: C, C++, Python

Web Technologies: HTML, CSS, JavaScript, React, Flask

Databases: MySQL, PostgreSQL, MongoDB

Libraries / Tools: NumPy, Pandas, Scikit-learn, Streamlit, Matplotlib, GitHub.

PROJECTS

MiniSTL GitHub

C++ · OOP · Templates · RAII · Iterator Design · HashMap

- Architected a **header-only, template-based C++ library** from first principles, implementing type-safe Vector, Stack, Queue, Linked List, and HashMap using abstraction, encapsulation, and operator overloading.
- Engineered mutable and const **iterator abstraction layers** decoupling algorithms from container internals, and implemented exponential capacity-growth with reserve()/shrink_to_fit() for amortized O(1) insertion.
- Designed a HashMap with **O(1) average-case lookup**, strictly enforced **Rule of Five with RAII** for exception safety, and built a Makefile-driven test suite for one-command verification.

PrepXP GitHub | Live Demo

React · FastAPI · Gemini API · Codeforces API · Recharts · REST APIs

- Built and deployed a **full-stack AI-powered analytics platform** integrating the **Codeforces API** with **Google Gemini AI** to deliver personalized competitive programming insights through a scalable, service-oriented backend architecture built entirely with FastAPI and Python from the ground up for real users.
- Designed reusable analytics modules for contest history, rating progression, topic-wise practice distribution, and problem difficulty analysis, exposing **production-grade REST APIs** consumed by an interactive **React** and **Recharts** frontend dashboard for real-time analytics and progress tracking over time.
- Deployed and maintained the full application stack across **cloud platforms** (Render, Vercel), ensuring reliable **CI/CD-driven delivery**, seamless cross-service integration, and consistent production uptime.

UCP Analysis GitHub

Python · Pandas · NumPy · Scikit-learn · Matplotlib · Seaborn

- Built an **end-to-end machine learning pipeline** for used-car price prediction, transforming raw vehicle listing data into model-ready features through systematic **data cleaning, outlier handling, and categorical encoding** for significantly improved model accuracy and overall reliability across real datasets and use cases.

- Conducted **exploratory data analysis (EDA)** using Matplotlib and Seaborn to uncover pricing trends, feature correlations, and distribution patterns, guiding feature selection and preprocessing decisions throughout the end-to-end modeling pipeline and full development workflow from start to finish.
- Trained and benchmarked **multiple regression models** using evaluation metrics like **RMSE and R^2**, then performed **feature importance analysis** to interpret model behavior and identify the primary factors used-car resale value.

RELEVANT COURSES

Core Courses: Data Structures and Applications, Algorithms, Database Management Systems, Operating Systems, Theory of Computation, Discrete Maths for Computer Science, Introduction to Programming.

Math Courses: Calculus I, Calculus II, Probability and Random Variables, Intro to Statistics, Elementary Linear Algebra, Differential Equations, Complex Variables

External Courses: Supervised Machine Learning: Regression and Classification, Advanced Learning Algorithms **(DeepLearning.AI, Coursera).** Google Student Upskilling Launchpad **(Google).**

ACHIEVEMENTS & EXTRACURRICULARS

LeetCode: Max Rating -- **1829** | Top 7% Worldwide | Solved **680+** problems | Handle -- krixchait

Codeforces: Max Rating -- **1357** | Pupil | Solved **130+** problems | **24+** contests | Handle -- krixchait

NCC B Certificate, National Cadet Corps, IIT Hyderabad

State Level Vedic Mathematics Champion, AGON 2020 -- Viswam Edutech School Labs

POSITION OF RESPONSIBILITY

Student Representative, SUGC -- Senate Undergraduate Committee, IIT Hyderabad

- Represent **650+** third-year undergraduates as a liaison between the **CR Council, DUGC, and AAC**, shaping **Senate-level** decisions on curriculum, credit policy, and academic evaluation standards.

Class Representative, ES28 Batch, IIT Hyderabad

- Serve as the primary point of contact between the batch of **40 students** and **faculty**.

Web Coordinator, Prayas -- Rural Development Centre, IIT Hyderabad

Student Mentor, Sunshine -- The Counselling Cell, IIT Hyderabad